

The courier has lost the route

Learning intention

Trace blood through the heart, lungs and body and explain the heart's pumping job.

A blood token must visit lungs and body.

Can you rescue its route without skipping the heart?

Which system does the delivery?

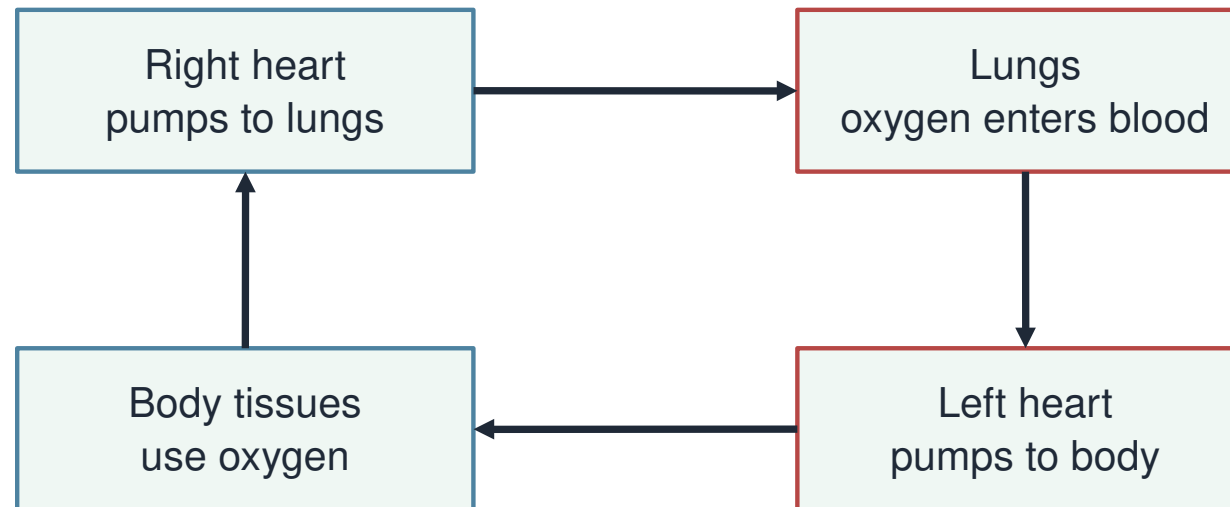
R1. What does the heart pump?

R2. What do blood vessels carry?

R3. Name a substance tissues need.

I trace one complete journey

One complete route: two heart sides

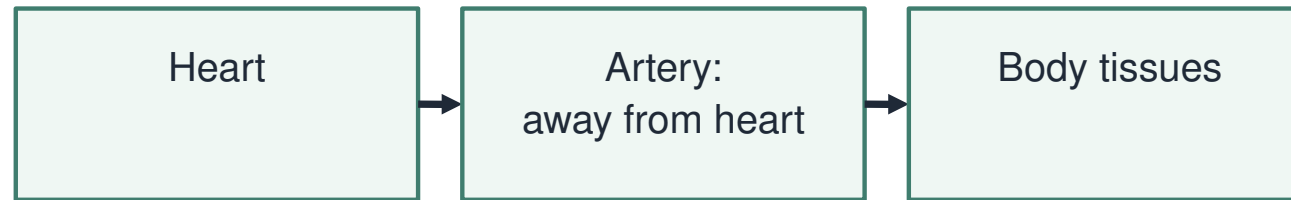


Clockwise. Colour marks oxygen content,
not the actual colour of blood.

Start at the right side of the heart. Visit lungs, then the left heart, then body. The heart is passed twice per complete circuit.

I name the vessel by direction

Name vessels by their direction

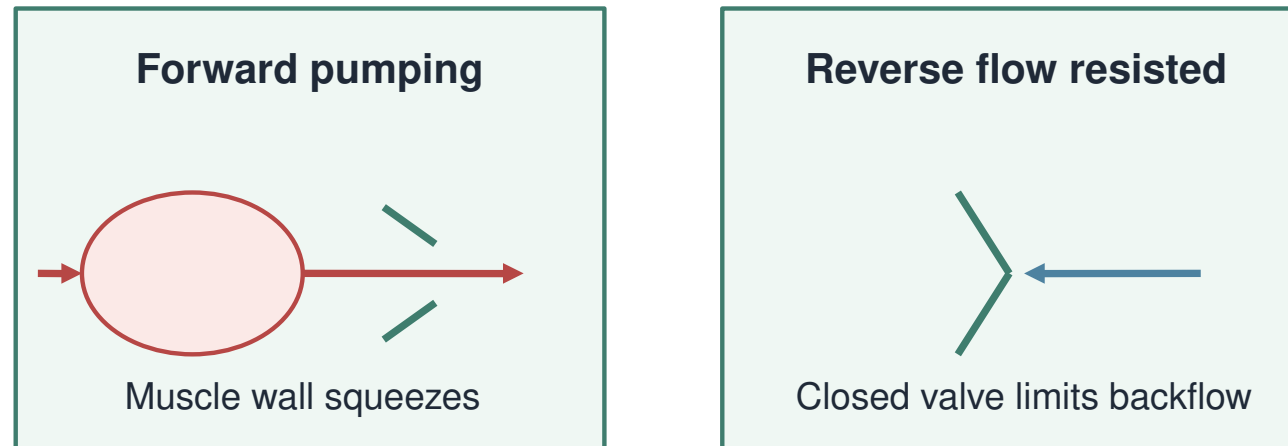


The return route uses veins towards the heart.
The heart-to-lungs artery also goes away.

An artery carries blood away from the heart. A vein carries blood towards the heart. Oxygen content does not define the name.

I explain the pump

Muscle pumps; valves guide the flow



Selective model: chamber shape and scale simplified.

Heart muscle contracts to move blood. Valves help stop backward flow. A valve guides flow; it does not power the pump.

Route rescue: four delivery links

Use the route clues on page 1. W1. Put A–D into a complete cycle. Start with A. Explain the next step.

Card	Delivery link
A	Right heart → lungs
B	Lungs → left heart
C	Left heart → body
D	Body → right heart

Find the two broken labels

Fault 1: “Lungs → body, skipping the heart.”

Fault 2: “All veins contain blue blood.”

W2. Repair each with a scientific reason.

An oxygen-poor artery?

W3. The vessel goes from heart to lungs.

Is it an artery or vein?

Decide using direction, then explain colour.

Blood leaves the lungs. Where next?

Choose the next main destination in this route.

A Straight to the body, bypassing the heart.

B To the left side of the heart.

C To the right side of the heart.

Build a reliable route map

Choose one response route.

Show the arrows and explain the pump.

Use the evidence sheet if needed.

Read your route aloud or silently

Can a token follow every arrow?

Does it return to where it started?

Have you passed through both heart sides?

Check the key

Write “oxygen-rich” and “oxygen-poor”.

Explain that blood itself is red.

Name an artery by direction.

Test another map

Trace one complete journey.

Identify one strong label.

Suggest one specific correction.

Choose a model improvement

Should the next version add capillaries or chambers?

Explain what that extra detail would help show.

A route in three answers

E1. After the lungs, where does blood go?

E2. What makes a vessel an artery?

E3. What powers the movement of blood?